

Image Formation Models for Improving Image Quality in Autonomous Driving

Question: Explain how image formation models help in improving image quality in applications such as autonomous driving.

Concept

Image formation models explain how light reaches the camera. In fog or haze, scattering and atmospheric light reduce visibility. These models estimate these effects and restore clearer images through dehazing.

Autonomous Driving

Improved image quality helps detect lanes, vehicles, pedestrians, traffic signs and obstacles more accurately in poor weather, improving safety and navigation.

Dark Channel Prior

The code computes the dark channel, estimates atmospheric light, estimates the transmission map, reconstructs the scene radiance, and produces a dehazed image from a hazy input.

Python Code

Include the complete Python code from the question here.

Result

The output shows a clearer dehazed image, transmission map, and estimated atmospheric light, demonstrating improved visibility.